

Kathy Cheng

PHD CANDIDATE · HUMAN-COMPUTER INTERACTION · UNIVERSITY OF TORONTO

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Education

University of Toronto

Toronto, ON

PHD IN MECHANICAL & INDUSTRIAL ENGINEERING

2021 – 2026

- Dissertation: Improving collaborative CAD systems through human–AI interaction and software engineering insights.
- Co-advisors: Dr. Alison Olechowski (Mechanical & Industrial Engineering); Dr. Shurui Zhou (Computer Engineering).

University of Toronto

Toronto, ON

BASc IN MECHANICAL ENGINEERING

2021

- Minors: Advanced Manufacturing; Environmental Engineering
- Undergraduate thesis: *An Analysis of Collaborative Computer-Aided Design Assembly*. Supervised by Dr. Alison Olechowski.

Professional Experience

University of Toronto

Toronto, ON

HCI RESEARCHER

Sept 2021 – Present

- Designed and implemented an LLM-powered, multimodal agent using GPT-4V, Pyrender, and Blender to automate change summarization for CAD models.
- Built Python-based data collection and analysis pipelines to curate and analyze 16K+ product reviews.
- Designed and led qualitative studies (40+ professionals) to identify core collaboration and coordination challenges.
- Analyzed 33M+ interaction logs from 30+ designers to model user behavior and inform the design of adaptive AI assistance.

Autodesk Research

Toronto, ON

HCI RESEARCH SCIENTIST INTERN

May 2025 – Sept 2025

- Developed and prototyped 6 agentic GenAI assistance concepts for professional CAD workflows, emphasizing reasoning, explanation, and user control (Think Aloud project).
- Designed and conducted 30 user studies to evaluate human–AI interaction patterns among casual and professional users.
- Published a full paper accepted to CHI 2026 and delivered 2 invited talks to industry and research audiences.
- Collaborated with the HCI & Visualization team, working with Dr. Jo Vermeulen, Dr. Justin Matejka, and Dr. George Fitzmaurice.

Scotiabank

Toronto, ON

TECHNOLOGY CONSULTANT

May 2019 – May 2023

- Executed a COVID-19 technology roll-out, increasing the department’s remote work capacity from 30% to 98% within 20 days.
- Developed Excel-based automation to optimize recruitment, improving turnaround time by 30% and intake volume by 50%.
- Collaborated with designers, developers, and product managers to deliver solutions tailored to diverse user needs.
- Led the migration of 300+ users’ primary telecommunications provider from Blackberry to Apple within 2 months.

Publications

PEER-REVIEWED CONFERENCE PROCEEDINGS

- [C9] **Cheng, K.**, Vermeulen, J., Fitzmaurice, G., Matejka, J. Lost in Translation: The Value of Verbalizations in Interpreting 3D Computer-Aided Design Workflows. *Accepted to ACM Conference on Human Factors in Computing Systems (CHI)*. 2026.
- [C8] Deng, Y., Zhang, S., **Cheng, K.**, Olechowski, A., Zhou, S. Untangling the Timeline: Challenges and Opportunities in Supporting Version Control in Modern Computer-Aided Design. *Accepted to ACM Conference on Human Factors in Computing Systems (CHI)*. 2026.
- [C7] **Cheng, K.**, Olechowski, A., Zhou, S. It’s a Complete Haystack: Understanding Dependency Management Needs in Computer-Aided Design. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*. 2025. [View here](#).
- [C6] **Cheng, K.**, Zhou, S., Olechowski, A. “A Lot of Moving Parts”: A Case Study of Open-Source Hardware Design Collaboration in the Thingiverse Community. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*. 2024. [View here](#).

- [C5] Zhang, K., **Cheng, K.**, Olechowski, A. Quantitative CAD Archetype Framework Evaluation with Professional User Data. *ASME International Design Engineering Technical Conference (IDETC-CIE)*. 2024. View here.
- [C4] Zhang, K., **Cheng, K.**, Olechowski, A. Developing a CAD Personality Framework Based on User Data. *Computer-Aided Design Conference and Exhibition (CAD)*. 2024. View here.
- [C3] **Cheng, K.**, Cuvin, P., Olechowski, A., Zhou, S. User Perspectives on Branching in Computer-Aided Design. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*. 2023. View here.
- [C2] **Cheng, K.**, Davis, M., Zhang, X., Zhou, S., Olechowski, A. In the Age of Collaboration, the Computer-Aided Design Ecosystem is Behind: An Interview Study of Distributed CAD Practice. *Proceedings of the ACM on Human-Computer Interaction (CSCW)*. 2023. View here.
- [C1] **Cheng, K.**, Olechowski, A. Some (Team) Assembly Required: An Analysis of Collaborative Computer-Aided Design Assembly. *ASME International Design Engineering Technical Conference (IDETC-CIE)*. 2021. View here.

PEER-REVIEWED JOURNAL ARTICLES

- [J4] Asuzu, C., **Cheng, K.**, Olechowski, A. The Personas of Cloud CAD Collaboration: A Case Study of a Professional Design Team. *IEEE Transactions on Engineering Management*. 2024. View here.
- [J3] **Cheng, K.**, Olechowski, A. Analysis of Collaborative Assembly in Multi-User Computer-Aided Design. *Journal of Mechanical Design*, 146(3): 031701. 2024. View here.
- [J2] Roy, D., Calpin, N., **Cheng, K.**, Olechowski, A., Arguelles, A., Zurita, N., Menold, J. Designing Together: Exploring Collaborative Dynamics of Multi-Objective Design Problems in Virtual Environments. *Journal of Mechanical Design*, 146(3): 031702. 2024. View here.
- 🏆 JMD Editor's Choice Award 2024
- [J1] Ferguson, S., **Cheng, K.**, Adolphe, L., Van de Zande, G., Wallace, D., Olechowski, A. Communication patterns in engineering enterprise social networks: an exploratory analysis using short text topic modelling. *Design Science* 8, e18. 2022. View here.

PEER-REVIEWED SHORT CONFERENCE PAPERS

- [S1] **Cheng, K.**, Zhang, Z., Deng, Y., Zhou, S., Olechowski, A. Fetch-A-Sketch: Navigating Dependency Density of 3D CAD Master Sketches. *Accepted to Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI)*. 2026.

IN REVIEW

- Velikonja, V., **Cheng, K.**, Olechowski, A. Exploring the Prevalence and Cause of Manufacturing Fixation in Design in Novice Engineering Designs via Computer-Aided Design. *Under review for Advanced Engineering Informatics*. 2026.

Awards, Fellowships, & Grants

2026	SGS Conference Travel Grant , University of Toronto	\$ 1690
	MIE Conference Travel Grant , University of Toronto	\$ 500
2025	SGS Conference Travel Grant , University of Toronto	\$ 690
2024	Canada Graduate Scholarship – Doctoral , NSERC	\$ 118,333
	Ontario Graduate Scholarship , Ontario Student Assistance Program, <i>Declined</i>	\$ 15,000
2023	Ontario Graduate Scholarship , Ontario Student Assistance Program	\$ 15,000
	MIE Conference Travel Grant , University of Toronto	\$ 500
	BPart Travel Award , American Society of Mechanical Engineers (ASME)	\$ 1,250
	1st Place Poster Presentation – MIE Graduate Research Symposium , University of Toronto	\$ 500
2022	Ontario Graduate Scholarship , Ontario Student Assistance Program	\$ 15,000
	Best Poster Design – Onshape Research Symposium , PTC Inc.	\$ 350
2021	Best Poster Award – Undergraduate Engineering Research Day , University of Toronto	\$ 100
	PEY Co-op Student of the Year Award , University of Toronto	–
2016	Dean's Merit Entrance Scholarship , University of Toronto	\$ 7,500
	President's Entrance Scholarship , University of Toronto	\$ 2,000

Presentations

* *presenting author*; + *mentored undergraduate*

INVITED TALKS

- [T6] Aug 2025. *Lost in Translation: The Value of Verbalizations in Interpreting 3D CAD Workflows*. Invited talk: Autodesk Research, Toronto, ON.
- [T5] July 2025. *Talking While Designing: The Value of Verbalizations in Interpreting 3D CAD Workflows*. Invited talk: Autodesk Research, Toronto, ON.
- [T4] Nov 2024. *Open-Source Hardware Design Collaboration*. Invited talk: Human-Centered AI Reading Group, McGill University, Online.
- [T3] Aug 2024. *The Trove of CAD Informatics: Acquiring and Analyzing CAD Data for Design Process Insights and AI Applications*. Workshop talk: ASME International Design Engineering Technical Conference, Washington D.C., USA.
- [T2] July 2024. *Open-Source Hardware Design Collaboration in the Thingiverse Community*. Invited talk: Machine Agency Reading Group, University of Washington, Online.
- [T1] May 2021. *Reflections on leadership skills and organizational considerations for the workplace of tomorrow*. Invited talk: 11th Conference on the Leader Engineer, Toronto, ON.

CONFERENCE PRESENTATIONS WITHOUT PROCEEDINGS

- [P13] **Cheng, K.***, Zhou, S., Olechowski, A. A Case Study of Open-Source Hardware Design Collaboration. Poster: University of Toronto MIE Graduate Research Symposium, Toronto, ON. 2024.
- [P12] **Cheng, K.***, Zhou, S., Olechowski, A. Analysis of Collaborative Software Development Insights to Physical Product Design via Computer-Aided Design. Poster: ASME (American Society of Mechanical Engineers) IDETC (International Design Engineering Technical Conference), Boston, MA. 2023.
- [P11] **Cheng, K.***, Zhou, S., Olechowski, A. Is Cloud the Future of Computer-Aided Design? An Industry Case Study of CAD Collaboration. Poster: Onshape Research Symposium, Virtual, Online. 2023.
- [P10] Roy, D.*, **Cheng, K.**, Olechowski, A., Menold, J. Exploring Collaborative Dynamics for Multi-Objective Design Problem Solving. Poster: Onshape Research Symposium, Virtual, Online. 2023.
🏆 Best Poster Award in the Informatics category
- [P9] Velikonja, V.** , **Cheng, K.**, Olechowski, A. Exploring the Prevalence and Cause of Manufacturing Fixation in Design (MFD) in Novice Engineering Designers via Computer-Aided Design (CAD). Poster: Onshape Research Symposium, Virtual, Online. 2023.
- [P8] **Cheng, K.***, Olechowski, A., Zhou, S. User Perspectives on Branching in Computer-Aided Design. Poster: University of Toronto MIE Graduate Research Symposium, Toronto, ON. 2023.
🏆 1st place in the Applied Mechanics & Design category
- [P7] **Cheng, K.***, Olechowski, A., Zhou, S. Time to branch out: An analysis of online user forum posts to inform Computer-Aided Design (CAD) branching. Poster: Onshape Research Symposium, Virtual, Online. 2022.
🏆 Best Poster Design Award
- [P6] Cuvin, P.** , **Cheng, K.**, Zhou, S., Olechowski, A. Where to Grow from Here? An Empirical Study of Branching Use in Computer-Aided Design. Poster: Onshape Research Symposium, Virtual, Online. 2022.
- [P5] Cuvin, P.** , **Cheng, K.**, Zhou, S., Olechowski, A. Where to Grow from Here? An Empirical Study of Branching Use in Computer-Aided Design. Poster: University of Toronto Undergraduate Engineering Research Day, Toronto, ON. 2022.
- [P4] **Cheng, K.***, Olechowski, A. An Analysis of Collaborative Computer-Aided Design Assembly. Poster: PTC Digital Transformation in Education Summit, Virtual, Online. 2021.
- [P3] **Cheng, K.***, Olechowski, A. A Study of Collaborative Computer-Aided Design Assembly. Poster: University of Toronto Undergraduate Engineering Research Day, Toronto, ON. 2021.
🏆 Best Poster Award in the Advanced Manufacturing category
- [P2] Davis, M.** , Zhang, X.+ , **Cheng, K.**, Zhou, S., Olechowski, A. What's Wrong with CAD?: Identifying and Classifying Challenges in Collaborative Work with Computer-Aided Design. Poster: University of Toronto Undergraduate Engineering Research Day, Toronto, ON. 2021.
🏆 Best Poster Award in the Transdisciplinary Engineering Education & Practices category

[P1] Zhang, X.⁺, Davis, M.⁺, **Cheng, K.**, Zhou, S., Olechowski, A. Challenges of Collaboration with Computer-Aided Design (CAD). Oral presentation: University of Toronto Undergraduate Engineering Research Day, Toronto, ON. 2021.

Teaching Experience

2024-2025 **MIE221: Manufacturing Engineering**, Head Teaching Assistant
2024-2025 **MIE221: Manufacturing Engineering**, Lab Manager
2024-2025 **MIE221: Manufacturing Engineering**, Marking Teaching Assistant
2025 **MIE221: Manufacturing Engineering**, Tutorial Teaching Assistant
2022-2023 **MIE221: Manufacturing Engineering**, Lab Teaching Assistant
2022-2024 **MIE301: Kinematics & Dynamics of Machines**, Lead Project Teaching Assistant
2021 **MIE301: Kinematics & Dynamics of Machines**, Project Teaching Assistant

Mentoring

2025 **Sally Zhang**, Computer Science undergraduate summer student.
2024-2025 **Felix Deng**, Mechanical & Industrial Engineering MASc student.
2023-2024 **Kevin Zhang**, Mechanical Engineering undergraduate thesis student.
2022-2023 **Victoria Velikonja**, Mechanical Engineering undergraduate thesis student.
2022 **Phil Cuvin**, Mechanical Engineering undergraduate summer student.
2021 **Michal Davis**, Engineering Science undergraduate summer student.
2021 **Jasmine Zhang**, Engineering Science undergraduate summer student.

Outreach & Professional Development

PEER REVIEW

2026. Reviewer for ACM Conference on Human Factors in Computing Systems (CHI).
2023-2025. Reviewer for ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW).
2023-2024. Reviewer for ASME International Design Engineering Technical Conference (IDETC-CIE).

VOLUNTEERING

2025. Student volunteer for IEEE/ACM International Conference on Software Engineering (ICSE).

PROFESSIONAL MEMBERSHIPS

2022-2025. Association for Computing Machinery (ACM) Student Member.
2021-2025. American Society of Mechanical Engineers (ASME) Student Member.

Skills

Programming: Python; R; VBA; MATLAB

AI & ML: Large Language Models (GPT-4/4V), prompt engineering, multimodal AI, agent-based systems

Data & Evaluation: Experimental design, qualitative coding, log analysis, mixed-methods evaluation

Other Software: Overleaf; Qualtrics; SurveyMonkey; Figma; NVivo; Miro; Jupyter Notebook; Dovetail; Camtasia; Zotero

Languages: English (fluent); Mandarin (advanced); French (intermediate)